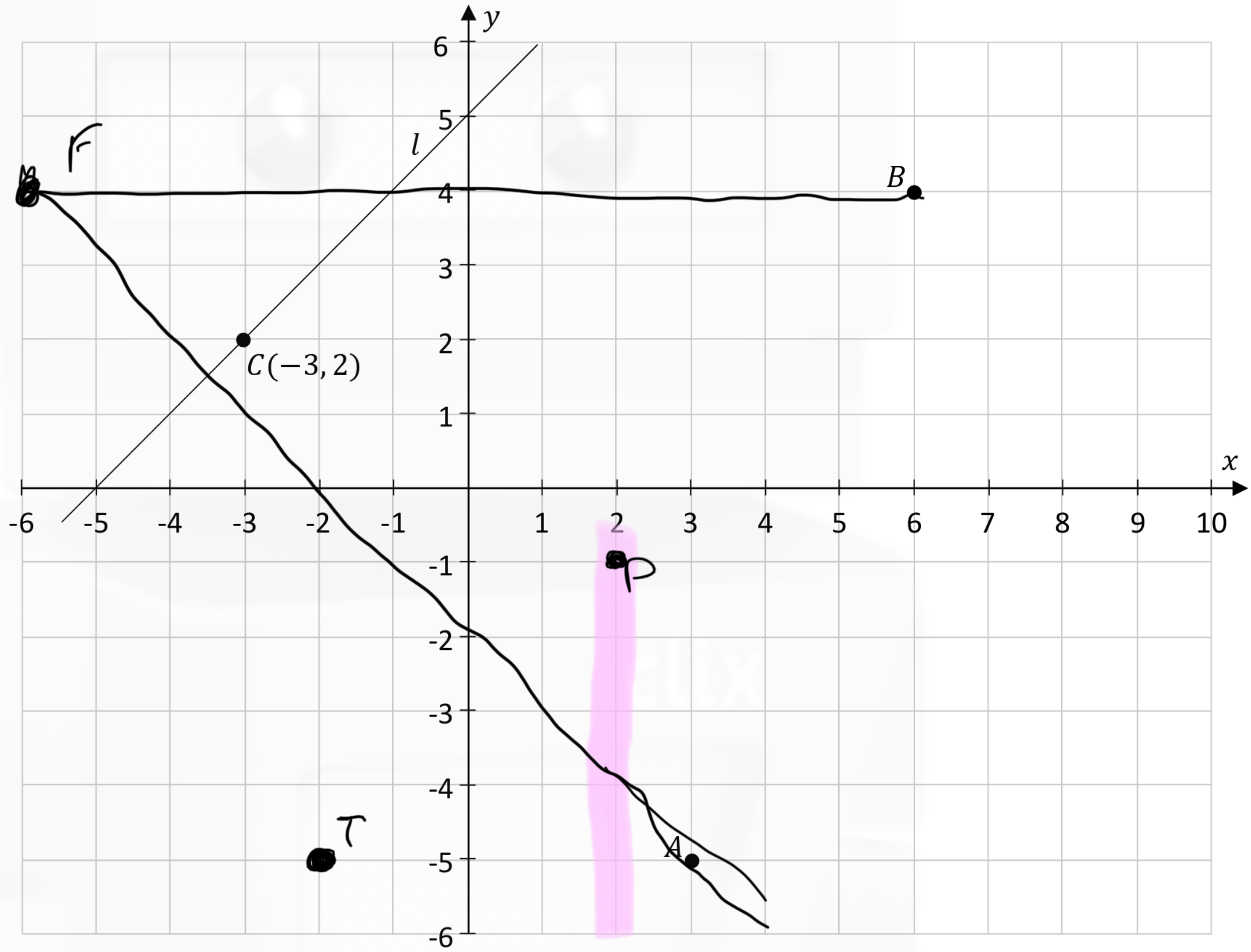
Question 10 **(50 marks)**

A TY maths class has created a game involving a co-ordinate treasure map, as shown below.
The game consists of tasks that involve directions, distances, and locations.
The tasks in the game are based on the map.



(a) **(i)** Treasure is hidden at location $T(-2,-5)$.
Mark T on the map where this treasure can be found.

(ii) Food is located at a point F on the map.
The point F is on a line which contains the point $A(3,-5)$ and has a slope of -1.
The point F is also on a line which contains $B(6,4)$ and has a slope of 0.
By drawing appropriate lines on the map above, or otherwise, find the co-ordinates of F.

(b) **(i)** A clue to another treasure is hidden in a locked box at point $B(6, 4)$.
The 4-digit code to open the box is d^4, where d is the distance from B to $C(-3, 2)$, and $d \in \mathbb{N}$.
Find the 4-digit code (d^4).

$$\sqrt{(-3-6)^2 + (2-4)^2}$$
$$\sqrt{(-9)^2 + (-2)^2}$$
$$\sqrt{81+4}$$
$$\sqrt{85}$$
7225

(ii) A meeting point $P(2, m)$ is below the x-axis.
P is a distance of $\sqrt{41}$ from point $B(6, 4)$.
Find the value of m **and** plot P on the map above.

$$\sqrt{(6-2)^2 + (4-m)^2}$$
$$\sqrt{(4)^2 +}$$
$$\sqrt{16+16-8m+m^2} = \sqrt{41}$$

$16 - 4m + m^2 - 4m$
$(4-m)(4-m)$

$$16+16-8m+m^2 = \cancel{41}$$
$$m^2 - 8m - 9 = 0$$
$$m = \frac{-(\cancel{8}) \pm \sqrt{-8^2 - 4(1)(-9)}}{2(1)} \quad \frac{8 \pm \sqrt{64+36}}{2}$$

$$m = \frac{9}{-1}$$
$$m = -1$$

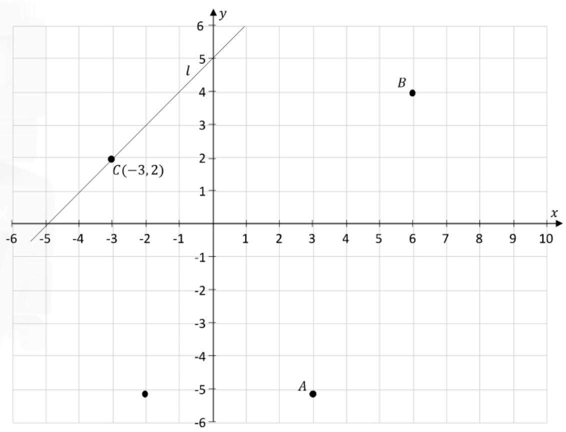
(c) **(i)** The line k has equation $x - y - 3 = 0$.
Verify, using substitution, that the point $T(-2, -5)$ is on k.

(ii) Another treasure also needs be somewhere on the line k.
You must pick a spot along k to contain this treasure.
Use algebra to find another point on k, other than T.

(iii) A spade for digging is hidden on line l which is parallel to k.
The line l contains the point $C(-3, 2)$.
Find the equation of line l.

Marking Scheme

(a)
(i)



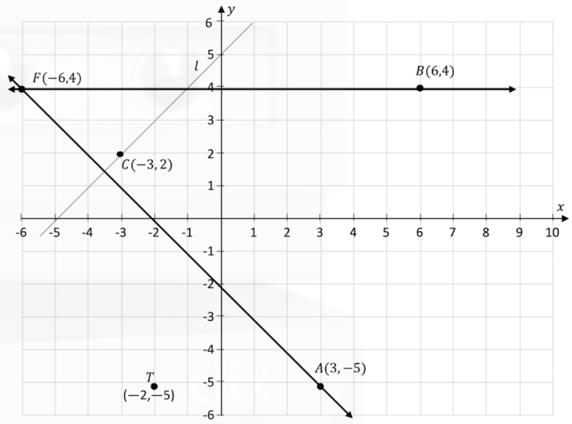
Scale 10B (0, 5, 10)

Partial Credit:

- Point plotted as $(-5, -2)$
- One or both coordinates plotted as a + value

(a) (ii)

$F(-6, 4)$



or

$$y-(-5) = -1(x-3)$$
$$x+y = -2$$

$$y-4 = 0(x-6)$$
$$y = 4$$

$$\Rightarrow x+4 = -2$$
$$x = -6$$

$$F = (-6, 4)$$

Scale 5C (0, 2, 3, 5)

Low Partial Credit:

Work of merit

- Slope = $\frac{\text{rise}}{\text{run}}$
- A or B labelled correctly
- Equation of line with some correct substitution
- One line correctly drawn through A or B

High Partial Credit:

Significant Work

- Both lines correctly drawn, F not identified
- Both equations found but not solved

(b) (i)

$$d = \sqrt{81+4}$$
$$= \sqrt{85}$$
$$= 9 (\in \mathbb{N})$$
$$\Longrightarrow d^4 = 6561$$

Note:

Accept $d = \sqrt{85},\ \ d^4 = 7225$ for Full Credit

Scale 10C (0, 3, 7, 10)

Low Partial Credit:

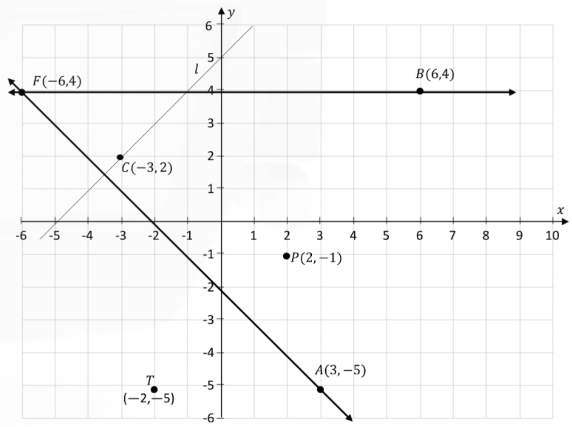
Work of merit

- Some correct substitution into formula to find d

High Partial Credit:

Significant Work

- Formula fully substituted
- Correct answer without work

(b) **(ii)**	$\lvert PB\rvert = \sqrt{16 + (m-4)^2} = \sqrt{41}$ $16 + m^2 - 8m + 16 = 41$ $m^2 - 8m - 9 = 0$ $(m+1)(m-9) = 0$ $\Rightarrow m = -1$ $(m = 9 \text{ not valid})$ $P(2, -1)$ (plot on map)  	**Scale 5D (0, 2, 3, 4, 5)** *Low Partial Credit:* Work of merit • Some substitution into Pythagoras or distance formula to find $\lvert PB\rvert$ *Mid Partial Credit:* • Formula fully substituted • Incorrect substitution into formula followed by consistent value of m *High Partial Credit:* Significant Work • Works to the quadratic equation • One relevant value of m identified but P not plotted **Note:** If 2 values of m given but only 1 point plotted award Full Credit
(c) **(i)**	$k: x - y - 3 = 0$ $T \in k \Longrightarrow (-2) - (-5) - 3 = 0$ $-2 + 5 - 3 = 0$ $[5 - 5 = 0]$ $[0 = 0 \text{ True}]$	**Scale 5C (0, 2, 3, 5)** *Low Partial Credit:* Work of merit • Some correct substitution into equation of line k *High Partial Credit:* Significant Work • Full correct substitution into equation of line k

(c) (ii)

Example:

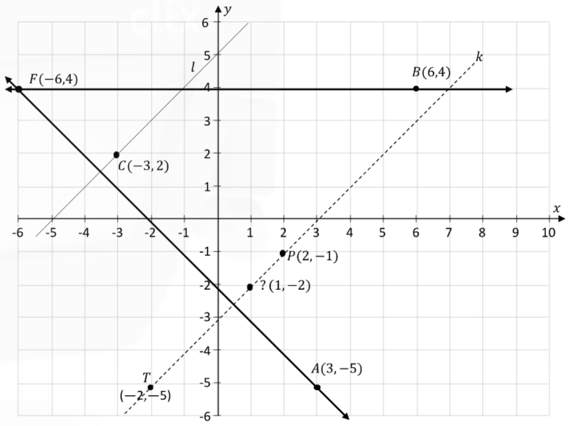
$$k: x - y - 3 = 0$$

$$Let\ x = 1$$

$$\Rightarrow 1 - y - 3 = 0$$

$$y = -2$$

$$(1, -2) \in k$$



Scale 5C (0, 2, 3, 5)

Low Partial Credit:

Work of merit

- Random selection of a domain value to find a point $(x, y) \in k$
- Graphical solution

High Partial Credit:

Significant Work

- Correct calculation of the corresponding range value

Note: Correct substitution of any point that is shown to satisfy the equation of line k merits Full Credit

(c) (iii)

$$k: x - y - 3 = 0$$

$$y = x - 3$$

$$\Rightarrow m_l = 1$$

$$\Rightarrow m_k = 1$$

Eqn of l:

$$y - 2 = 1[x - (-3)]$$

$$y - 2 = x + 3$$

or

$$y = x + 5$$

or

$$l: x - y + 5 = 0$$

Scale10C (0, 3, 7, 10)

Low Partial Credit:

Work of merit

- Effort at finding slope of k or l
- Relevant work on diagram
- Slope = $\frac{\text{rise}}{\text{run}}$

High Partial Credit:

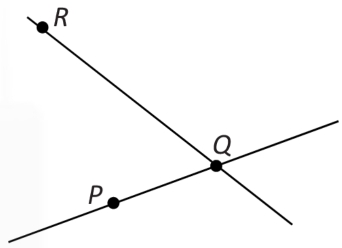
Significant Work

- Equation of line formula with full substitution

Note: Accept $y - 2 = 1(x + 3)$ for Full Credit

Question 2 **(25 marks)**

The diagram shows the line PQ and the line QR.
The co-ordinates of the points are $P(4, 2)$, $Q(8, 5)$ and $R(2, 11)$.



(a) Find the slope of PQ.

(b) Find the equation of the line PQ.
Give your answer in the form $ax + by + c = 0$, where $a, b, c \in \mathbb{Z}$.

(c) Write down the slope of any line perpendicular to *PQ*.

Slope =

(d) Find the area of the triangle *PQR*.

(a)	$$m = \frac{y_2 - y_1}{x_2 - x_1}$$ $$= \frac{5-2}{8-4}$$ $$= \frac{3}{4}$$	**Scale 10C(0, 4, 6, 10)** *Low Partial Credit:* • Work of merit • Formula with some correct substitution *High Partial Credit:* • Formula substituted correctly *Full Credit:* • Correct answer without work *Misreading:* • Finds slope of PR or QR
(b)	$$y - y_1 = m(x - x_1)$$ $$y - 2 = \frac{3}{4}(x - 4)$$ $$3x - 4y - 4 = 0$$	**Scale 5C (0, 2, 3, 5)** *Low Partial Credit:* • Formula with some correct or consistent substitution *High Partial Credit:* • Formula substituted correctly or consistently • One incorrect substitution followed by correct solution • Answer not in the required format **Note:** Accept $-3x + 4y + 4 = 0$
(c)	$$-\frac{4}{3}$$	**Scale 5B (0, 2, 5)** *Mid Partial Credit:* • Work of merit *Full Credit:* • Correct answer without work

(d)	$(4,2)\ (8,5)\ (2,11)$ $\downarrow \quad \downarrow \quad \downarrow$ $(0,0)\ (4,3)\ (-2,9)$ $=\frac{1}{2}\lvert x_1y_2 - x_2y_1\rvert$ $=\frac{1}{2}\lvert(4)(9)-(-2)(3)\rvert$ $=\frac{1}{2}\lvert 36+6\rvert$ $= 21$ square units	**Scale 5D (0, 2, 3, 4, 5)** *Low Partial Credit:* • Translation of any relevant point to (0,0) • Area of triangle formula with some substitution *Mid Partial Credit:* • Area of triangle formula substituted fully with 2 from P,Q, and R *High Partial Credit:* • Formula substituted correctly with translated points • Area of triangle found from formula substituted fully with 2 from *P,Q, R* • An incorrect translation applied correctly

Question 8 **(40 marks)**

The diagram below shows the triangles *CBA* and *CDE*.

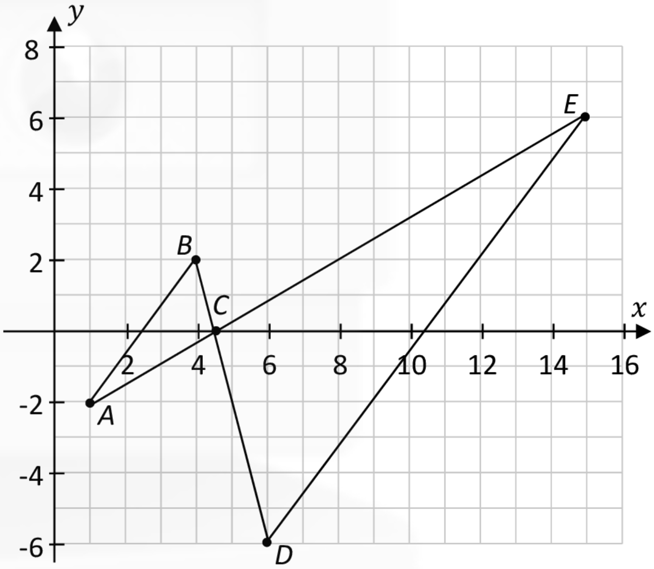
(a) The coordinates of *C* are $(4{\cdot}5, 0)$.
From the diagram, write down the coordinates of the points *A*, *B*, *D*, and *E*.

A = (1 , −2)

B = (4 , 2)

D = (6 , −6)

E = (14 , 6)



(b) Show, using slopes, that the line segments [*AB*] and [*DE*] are parallel.

$$\frac{y_2 - y_1}{x_2 - x_1} \Rightarrow \frac{2 - -2}{4 - 1} = \frac{4}{3}$$

$m_{AB} = \frac{4}{3}$

$m_{DE} = \frac{4}{3}$ $\frac{6 - -6}{14 - 6} = \frac{12}{9} = \frac{4}{3}$

(c) **(i)** Show that the area of the triangle *CBA* is 4 square units.

(1, −2) (4, 2) (4.5, 0)

(0, 0) (3, 4) (3.5, 2)

½ [

(ii) Find $|AB|$, the distance from *A* to *B*.

(iii) The triangle *CDE* is an enlargement of the triangle *CBA*.
Given that $|DE| = 15$ units, find the scale factor of the enlargement.

Scale factor =

(iv) Use this scale factor to find the area of the triangle *CDE*.

(a)	$A = (1, -2)$ $B = (4, 2)$ $D = (6, -6)$ $E = (15, 6)$	**Scale 10C (0, 3, 7, 10)** *Low Partial Credit:* • <u>ONE</u> correct point. *High Partial Credit:* • <u>THREE</u> correct points. • All answers written consistently as $(y,\ x)$
(b)	$m_{AB} = \frac{4}{3}$ $m_{DE} = \frac{12}{9}$ **or** $\frac{4}{3}$ **Conclusion:** Parallel, or slopes are the same.	**Scale 10D (0, 4, 5, 7, 10)** *Low Partial Credit:* • Any mention of same slopes or similar. • Slope formula written with some relevant substitution. • Rise over Run (Written or indicated). *Mid Partial Credit:* • Slope formula fully substituted for one slope. • One correct slope. *High Partial Credit:* • Both slopes but no conclusion. • One error in substitution followed by correct calculation and consistent conclusion. *Full credit:* • Equal slopes and conclusion.
(c) **(i)**	$A = (1, -2)\ B = (4, 2)\ C = (4{\cdot}5, 0)$ $\downarrow \quad \downarrow \quad \downarrow$ $A = (0, 0)\ B = (3, 4)\ C = (3{\cdot}5, 2)$ $\text{Area} = \frac{1}{2}[(3)(2) - (3{\cdot}5)(4)]$ $\text{Area} = \frac{1}{2}\lvert(6) - (14)\rvert$ $= \frac{1}{2}\lvert(-8)\rvert$ $= 4$	**Scale 5C (0, 2, 3, 5)** *Low Partial Credit:* • Correct relevant formula with some correct/consistent substitution. • One point translated to $(0, 0)$. *High Partial Credit:* • $\text{Area} = \frac{1}{2}\lvert x_1y_2 - x_2y_1\rvert$ fully substituted with translated points. • Error in translation and area calculated correctly. • No translation but uses TWO of the points A, B, C and area calculated correctly. • One error in substitution followed by correct calculation.

(c) (ii)	$\|AB\| = \sqrt{3^2 + 4^2} = 5$ units	**Scale 5C (0, 2, 3, 5)** *Low Partial Credit:* • Correct relevant formula with some substitution. • Any work of merit indicated in this part. *High Partial Credit:* • Correct substitution without calculation. • One error in substitution followed by correct calculation. • Correct answer without work.
(c) (iii)	$\frac{15}{5} = 3 = k$	**Scale 5B (0, 3, 5)** *Partial Credit:* • Use of answer to part (c)(ii). • Scale factor formula written or indicated. • Attempts to calculate $\|AB\|$ in this part. *Full Credit:* • Correct answer without work.
(c) (iv)	Area $= 4 \times 3^2 = 36$ units^2	**Scale 5C (0, 2, 3, 5)** *Low Partial Credit:* • Work of merit with answer from (c)(iii). • Work of merit with answer from (c)(i). *High Partial Credit:* • Area as $3 \times 4 = 12$ or equivalent. *Full Credit:* • Correct answer without work.